

StaleLine: appendix to the quant note (HYPOTHESIS_v4 edition)

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A Earlier pre-registered study (v3)

The v3 study (economic hypothesis, data, methodology and results, with the lag figure, the results table and the equity curve) is in the main note, section “Before the plot twist: the pre-registered v3 study (not v4)”. Nothing in it is v4; v4 has no result yet. This appendix keeps only the v3 detail that is not in the main note: the pre-registration commit list (Appendix B), the post-hoc record (Appendix C) and the Sharpe method (Appendix E).

B Disclosure (v3 holdout run, not v4)

v3 pre-registration commits on `main`, in order (ET). The holdout was evaluated once, from `872ff43`, 2026-10-04 02:00:23 ET to 02:32:24 ET. Data exclusions and fixes, recorder incidents and a stop-script failure handled by hand before the run are listed in `docs/holdout_run_disclosure.md`, together with every post-hoc disclosure (Idea 3 invalid, lookahead; Idea 6 winner leg invalid, ESPN end timestamps; Idea 11 original execution model invalid, conditioned on the future).

The next table lists v3 pre-registration commits; none of it is v4.

Commit	ET time	What
<code>f12c1dc</code>	2026-10-02 22:20	Retire the CME
<code>d6bfa9c</code>	2026-10-03 00:01	v2: Kalshi vs po
<code>bc9d583</code>	2026-10-03 00:41	v2 Amendment
<code>508f2db</code>	2026-10-03 03:25	v3: Strategy A (
<code>8a509ff</code>	2026-10-03 11:15	v2 Amendment
<code>9507d6a</code>	2026-10-03 11:42	v3 Amendment
		B fill corrections
<code>37170aa</code>	2026-10-03 13:57	v3 Amendment
<code>ed5db9b</code>	2026-10-03 18:14	v2 Amendment
		taker delay), pla
<code>8964ea9</code>	2026-10-03 19:30	Power note to v
<code>d349e50</code>	2026-10-03 19:40	v3 Amendment
		ference games ex
<code>f6aa3b5</code>	2026-10-03 20:25	v2 Amendment
		112).
<code>cf8f4e9</code>	2026-10-03 21:22	v2 Amendment
		bootstrap CI.
<code>9f469e7</code>	2026-10-03 23:20	v3 Amendment
		3); complete file
<code>04c3843</code>	2026-10-03 23:45	v3 Amendment
		matches vs ESP

C Post-hoc work on the v3 data (not v4)

After the holdout run we tried 12 further numbered ideas, a Polymarket US time-and-sales test and a training-positive book, each with a specification committed before any real-data code ran. All

are labeled post-hoc and exploratory and none changes a pre-registered result. Ideas run on holdout-period recordings have no separate holdout. In total 77 variants are logged in `experiments/variants.csv` and used in the deflated Sharpe. Later work, all post-hoc and on training data unless stated, is logged in `results/posthoc/LEDGER.md` on main: a walk-forward meta-model (2 variants, 07738c9) and a context map (2, e3da0d5); a broad search (54 trials, Reality Check p 0.99, SPA p 0.621); cost-side rounds (180), maker (36) and pattern (57) searches; unsupervised regimes (64); residual regression (48); lead-lag rounds (40); a frozen Laya encoder with a ridge head (2); a Kalshi-signal market-making test on the slow venue (4 variants, 8cb29f7; verdict “does not make sense on this data”); a Llama LoRA test stopped with 0 of 3 trials evaluated. That is 485 further trials, 566 in total. White’s Reality Check over the largest combined set (479 trials) gives $p = 0.47$; no subset is below 0.42. The plain maker that never pulls quotes (the baseline of that test) earned 1.09 cents per contract at +60 s [0.17, 2.12] on polymarket.com, but \$41.75 of its \$46.53 came from five volatile games, it is 0.15 cents without them, matched round trips earned 0.52 cents per contract, quoting at a 1-cent spread lost -0.62 cents, and its settlement result spans zero [-0.62, 6.06]; under a reading rule written after these results were seen, all four conditions pass on the 59 games with fills, so we treat it as a lead that needs more testing on live games (over all 88 games the game-share condition fails). HYPOTHESIS_v4 (94aa1a0) was committed before its test data was opened; its test set 1 could not run, because all 6 late Oct 3 games have recordings that end at 00:34 ET, before the games ended. On those truncated recordings, outside the v4 rule and not evidence either way, arm A made \$2.76 on 34 fills and arm B \$2.45 on 7 fills at +60 s. The confirmatory test is the Oct 8 to 12 games. Descriptive only (no trials): Kalshi versus CME on 14 playoff games with CME book data (v3-era, not v4: 18 taker settings, all negative after CME’s spread; a resting-order variant, also negative), a broker and venue survey, and news and sportsbook-line collection (no usable intraday timing). Databento spend for this work: \$0.88. Trade-based lead-lag measures in these rounds are not used as findings, because trade-based lead-lag is biased toward the busier venue (main note, v3 section, lag figure).

Idea	Hypothesis	Spec (ET)	Var.	Head
T&S lag- gard	PM US prints lead Kalshi	a510ba0 03:33:08	3	net [-4.
1	Kalshi complement mispricing	d32fa2c 03:35:33	3	L 0 \$/t -0.2
2	polymarket.com leads sub-second	4ec43cc 03:33:00	3	L 0 -11.
3	fade large Kalshi prints	4e60e8c 03:45:47	3	not
4	late-game favorite underpriced	3142f9b 03:51:37	4	RO +0. -1.6
6	buy winner after end marker	7c78c2c 03:57:49	3	not leg
7	pregame cross-venue gap	a413026 04:38:20	3	non tra

Every number in the next table is v3-era post-hoc work, not v4.

8	fade pregame taker flow	856b5f9 04:05:34	3	RO -0.0 c/c
9	ESPN win prob. vs Kalshi	df80d2c 04:03:47	4	RO +0. -0.9
10	fade scoring-play moves	3a02690 04:03:49	3	net [-4.
11	lag and cross-venue arbitrage	277769a 04:11:44	3	cor nak \$-3
12	in-game cross-venue gap	ca72fb9 04:38:41	3	RO +0. -1.1
13	NFL closing line as fair value	03ab9a5 04:38:36	3	non tra

D HYPOTHESIS_v4: the live hypothesis

HYPOTHESIS_v4 is the project’s live hypothesis. It is untested: no confirmatory result exists yet. It was committed to `main` as `HYPOTHESIS_v4.md` in `94aa1a0` at 07:56:29 ET on 2026-10-04, before any of its test data was examined. Its Arms A and B are two settings of one passive maker; they are unrelated to the v3 strategies A and B (main note, section “Before the plot twist”). The frozen text, verbatim:

HYPOTHESIS_v4 (post-hoc origin, frozen before test data is examined).

Origin: posthoc-mm-signal N maker, polymarket.com, Oct 3 games kicking off by 20:00 ET (863eb69, f535256): +1.09 c per contract at +60 s [0.17, 2.12], concentrated in 5 games, matched

round trips +0.52 c.

Rule: passive maker quoting both sides at the best bid and ask (join, never improve), 10 contracts, inventory cap +/- 50, no Kalshi signal. Arm A: always quote. Arm B: quote only while the spread is ≥ 3 c.

Venues: polymarket.com (recorded books and trades); Polymarket US (authenticated order-book websocket with venue timestamps, recorded live; test set 2 only).

Fills: strict trade-through only. Maker fee: each venue's published schedule at the time (cite).

Kickoff cut [T - 2 min, T + 20 min].

Primary metric: P&L per contract marked to mid at +60 s, game-bootstrap 95% CI (2,000 reps, seed 20261011). Secondary: matched round-trip P&L; held to settlement; excluding the top 5 games.

Test set 1 (run once now, descriptive only, small n): Oct 3 games kicking off after 20:00 ET with complete polymarket.com books and trades on disk, never used by posthoc-mm-signal.

Test set 2 (confirmatory): all college and NFL games from 2026-10-08 00:00 ET to 2026-10-13 00:00 ET, run once after the window closes.

Decision (test set 2, per venue and arm, Holm across venues): supported only if (1) primary CI above zero, (2) point estimate above zero excluding the top 5 games, (3) settlement P&L point estimate above zero. Otherwise not supported. Report either way.

Status. Test set 1 was not run: 0 eligible games, because every late Oct 3 recording ends at 00:34 ET, before those games ended (`posthoc-mm-v4 1d3d68a`). Test set 2 (Oct 8 to 13) needs live recording of polymarket.com books and trades and the Polymarket US order book with venue timestamps; that recorder is not yet running. The sealed Aug to Oct 2 holdout cannot test v4: for that period the repository holds only polymarket.com taker trades at 1 s resolution and no order books, so the join and trade-through fill rule cannot be simulated.

Where v4 came from (descriptive, one day, not pre-registered). A simulation on recorded Oct 3 polymarket.com top-of-book data (88 games, 428 fills, maker fee 0, queue position unknown) earned \$46.53 marked to mid at +60 s and \$114.03 held to settlement (`posthoc-mm-signal b346230, 863eb69`). Excluding the top 5 games it is +0.15 c per contract $[-0.48, +0.77]$, and 57.6% of the 59 games with fills were positive (38.6% of all 88). Spread captured was +\$63.18 against $-\$15.65$ of adverse selection. Under the reading rule applied to the 59 games with fills, all four conditions pass, so the team reads it as a lead worth testing live (`863eb69`); the rule was written after these numbers were seen, and read over all 88 games (38.6% positive) its game-share condition fails, which is why the live test decides. The 88 games are holdout-period games (Oct 3 and 4); no committed file records an authorization to use them, and we disclose it here.

E Sharpe ratios: method

For the v3 strategies, `report_book.py` sums net P&L per ET calendar day over the 60 holdout days (2026-08-06 to 2026-10-04), keeping zero days, divides by a fixed capital base (which does not change the Sharpe), and reports mean over standard deviation times $\sqrt{365}$, with a risk-free rate of 0; the NW5 version uses a Newey-West 5-lag standard deviation. We recomputed these from the committed trade files (`results/holdout/, main f4f3e28`) and they match `numbers.json`. 95% intervals from a 5-day block bootstrap (10,000 draws), v3 strategies only (not v4): v3 Strategy A $-3.81 [-6.05, -0.50]$, v3 Strategy B $-7.76 [-9.95, -6.07]$, v3 combined $-7.78 [-10.13, -5.97]$. Daily skew is about -3.2 to -3.7 , so the bootstrap intervals are the ones to quote. For v4's origin, only one day exists, so no daily or annualized Sharpe is possible; the per-game Sharpe (mean over standard deviation of game P&L, 88 games, game bootstrap) is +0.22 $[+0.06, +0.35]$ at +60 s and +0.16 $[-0.04, +0.35]$ held to settlement.